AI-Enhanced Coral Reef Habitat Mapping and Bleaching Detection around Kavaratti Island

Mini Project Report

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Title Page

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Certification

This is to certify that the project report titled “AI-Enhanced Coral Reef Habitat Mapping and Bleaching Detection around Kavaratti Island” has been carried out by Dheeraj, Niveditha S Nair, and Chithrabhanu under my supervision and guidance in partial fulfillment of the requirements for the award of the Degree of Bachelor of Technology in Computer Science and Engineering (Artificial Intelligence) of Chinmaya Vishwa Vidyapeeth, Kochi.

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Abstract

This project, titled “AI-Enhanced Coral Reef Habitat Mapping and Bleaching Detection around Kavaratti Island,” presents an integrated approach for automated monitoring of coral reef health using Artificial Intelligence (AI) and Computer Vision techniques. Coral bleaching, a critical environmental issue caused primarily by rising sea temperatures, threatens marine biodiversity. Our system combines deep learning–based object detection (YOLOv8) and HSV color-space analysis to identify and quantify coral bleaching from underwater videos. The YOLOv8 model segments coral regions with high accuracy, while HSV analysis cross-verifies color intensity for bleaching estimation. The results are visualized using charts and GIS-based spatial mapping. The project demonstrates the efficiency of AI-assisted monitoring for sustainable reef management and offers a framework adaptable for future large-scale marine studies.

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Chapter 1 – Introduction

Coral reefs are among the most vital and diverse ecosystems on Earth, yet they are increasingly threatened by ocean warming and environmental stressors. Coral bleaching occurs when corals expel the symbiotic algae (zooxanthellae) that give them color and nutrition, leading to whitening and possible mortality if prolonged. Monitoring and quantifying coral bleaching are crucial for effective conservation.

This project leverages Artificial Intelligence and image processing to automate the detection and analysis of coral bleaching around Kavaratti Island. Traditional manual observation methods are labor-intensive, subjective, and limited in scale. By integrating deep learning with color-based analysis, our framework provides an efficient, consistent, and scalable solution for coral reef health assessment.

Objectives:

- To automate coral bleaching detection using YOLOv8 segmentation.
- To cross-verify bleaching estimation using HSV color thresholding.
- To visualize bleaching distribution through GIS and statistical graphs.
- To build a prototype framework suitable for real-time reef monitoring.

Chapter 2 – Literature Review

Recent studies have emphasized the role of artificial intelligence, remote sensing, and deep learning in marine ecosystem monitoring. Research from 2021 onwards has shown a strong shift toward automation and interpretability.

Early works focused on governance and community management of reef ecosystems, while later studies adopted Convolutional Neural Networks (CNNs) for image-based coral classification. Between 2022 and 2024, remote sensing technologies using Sentinel-2 and PRISMA satellites allowed large-scale reef assessments. YOLO-based frameworks emerged as powerful real-time detection models capable of handling underwater conditions such as light scattering and turbidity.

Our project builds upon this progress, combining YOLOv8 segmentation and HSV pixel-level analysis for robust and interpretable bleaching detection. This dual approach balances model accuracy and visual interpretability, contributing to sustainable reef monitoring strategies.

Chapter 3 – System Design and Methodology

The system architecture integrates video-based coral imagery with AI-driven segmentation and color-space analysis. The major components are:

1. Data Acquisition – Underwater videos were collected around Kavaratti Island using diver-based recording. Frames were extracted at 1 fps using OpenCV.
2. Preprocessing – Extracted frames were organized into 'healthy' and 'bleached' folders. Segment Anything (SAM) was used for initial mask generation.
3. YOLOv8 Model Training – The YOLOv8s-seg model was fine-tuned on the dataset to classify coral regions into healthy and bleached categories. The best model weights were saved for inference.
4. HSV Cross-Verification – HSV-based color thresholding was implemented using cv2.inRange() to detect bleached regions based on hue, saturation, and brightness levels.
5. Result Computation – Bleaching percentage per frame was computed as (bleached_pixels / total_pixels) × 100.
6. Visualization – Results were stored in CSV format and displayed as pie charts and tables using Matplotlib.

Chapter 4 – Implementation and Technologies Used

The entire system was implemented in Python using Google Colab for cloud-based GPU training and inference.

Technologies:

- YOLOv8 (Ultralytics): Deep learning framework for object detection and segmentation.
- Segment Anything Model (SAM): Used for automatic coral mask generation.
- OpenCV: For frame extraction, preprocessing, and color-space analysis.
- NumPy & Pandas: For data computation and bleaching ratio tabulation.
- Matplotlib: For generating graphs and visualizations.
- scikit-learn: For implementing Gaussian Mixture Models (GMMs) during HSV classification.
- Google Drive Integration: Used for dataset, weights, and results storage.

Bug Fixing and Debugging:

During model training, issues such as unbalanced class labels and lighting noise were encountered. These were mitigated by augmenting data (rotation, brightness adjustment) and applying morphological operations to clean masks. GPU memory allocation errors were solved by reducing batch size and using mixed precision training. HSV sensitivity to lighting was minimized through adaptive thresholding and background filtering.

Chapter 5 – Results and Discussion

The YOLOv8 model achieved an accuracy of approximately 92% in classifying healthy and bleached coral regions. HSV-based pixel analysis achieved about 80% accuracy and served as a valuable cross-validation mechanism.

Outputs included per-frame bleaching percentages, averaged over each video. YOLOv8 proved robust to moderate lighting variations, while HSV analysis was faster and interpretable. Combining both approaches enhanced reliability.

Example results:
Video 1: 97.3% bleaching
Video 2: 98.0% bleaching
Video 19: 62.1% bleaching

Pie charts and CSV summaries provided clear insights into reef health status. The system demonstrated that integrating AI with color analytics can substantially improve accuracy and usability in marine monitoring applications.

Chapter 6 – Limitations and Future Scope

Limitations:

- Sensitivity to lighting and water clarity still affects segmentation accuracy.
- Dataset was limited to a single location (Kavaratti Island), reducing generalization.
- Single-time survey prevents long-term temporal analysis.
- YOLOv8 model requires GPU for efficient inference.

Future Scope:

- Expand dataset across multiple reef sites and time periods.
- Integrate satellite-based reef monitoring (Sentinel-2, PRISMA).
- Develop predictive models for reef health trends using time-series data.
- Deploy real-time web dashboard for live coral bleaching analytics.

Chapter 7 – Conclusion

The project successfully developed a dual AI–based framework combining YOLOv8 deep learning and HSV color-space analysis for automated coral bleaching detection around Kavaratti Island. The system achieved reliable detection accuracy and interpretability, providing visual and quantitative insights into reef health.

This project demonstrates how modern AI can assist marine conservation by transforming video footage into actionable data. The experience offered significant learning in data handling, model training, debugging, and interdisciplinary application of artificial intelligence. With future extensions, this framework could become a standard tool for coral reef monitoring and management worldwide.

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